

PAPER_1503 — GW Damping Factor BNS = $1/(D_{\text{phys}}-1) = 1/3$ EXACT

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket E (GW Events) **Date:** June 18, 2026 **Status:** CLOSED — BNS gravitational-wave total damping factor exact $1/3$

Observation

PAPER_011 (Stochastic Gravitational-Wave Background — UQFF Implications) derives the frequency-dependent damping factor for binary-neutron-star (BNS) GW events as $D_{\text{total}}(\text{BNS}) = 3.33 \times 10^{-1} \approx 0.333$, where $h_{\text{UQFF}} = D_{\text{total}} \cdot h_{\text{GR}}$ and $\Omega_{\text{GW,UQFF}} = D_{\text{total}}^2 \cdot \Omega_{\text{GW,GR}}$.

UQFF Closed Identity

$$D_{\text{total}}(\text{BNS}) = 1 / (D_{\text{phys}} - 1) = 1/3 = 0.333\ldots \quad \text{EXACT}$$

Physical Interpretation

BNS mergers radiate gravitational waves whose total transmitted amplitude through the SCm vacuum substrate is reduced to exactly one-third of the GR prediction. The factor $1/(D_{\text{phys}} - 1)$ is the same triad-symmetry denominator that yields: - Solar v_e deficit $1/3$ (PAPER_062) - SU(3) color charge 3 (PAPER_1373) - SCm carrier velocity $c/3$ (PAPER_1497) - TDE outflow factor $c \cdot 3/10$ (PAPER_1500)

The shared denominator unifies neutrino, gauge, jet, and GW phenomenology under a single integer-primitive ratio.

Predictive Implication

Squared damping $\Omega_{\text{GW}} = (1/3)^2 = 1/9 = 0.111$ transmits to the stochastic GW background — gives LIGO/Virgo-O5 a structural prediction independent of source-population fits.

NOT REPLACEMENT

GR predicts undamped propagation in vacuum. UQFF supplies a structural reduction at exactly $1/(D_{\text{phys}} - 1)$ due to triad-channel splitting in the SCm substrate.

Reference

- Source: PAPER_011 Stochastic GW Background UQFF Implications
- Related: PAPER_1504 (BBH damping), PAPER_1497 (SCm velocity), PAPER_1500 (TDE)
- Calculator dispatch: `calculate_paradox({"paradox": "gw_damping_bns_one_third"})`

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